

IT353 BLOCKCHAIN TECHNOLOGY



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Module - II

Fundamental concepts of Blockchain

- Concepts of Block, Transactions, Hashes, Consensus.
- Hashes: Hash cryptography, Encryption vs. hashing,
- Transactions: Recording transactions, Digital Signature, Verifying and confirming transactions
- Blocks and blockchain: Hash pointers, Blocks
- Consensus building. Distributed consensus, Byzantine generals problem
- Consensus mechanism: POW, POS, POB, POA, etc.
- Blockchain Architecture, Markle Root Tree.

Introduction to Blockchain

- Blockchain are often confused with cryptocurrencies specifically Bitcoins.
- **Blockchain** is a decentralized ledger, tracking digital assets on a Peer-to-Peer network.
- A **ledger** is a record book, which holds details of every transaction that takes place on the network.
- Each system has a copy of this ledger, hence this ledger is **distributed**.
- Central Database vs. Blockchain

Characteristics	Database	Blockchain
Data Ownership	Central authority	Users
Privacy and Security	Centrally managed	Cryptographic applications
Access Control	Centrally controlled	Same for all permissioned users
Trust	Trusted authority	Immutability
Reliability and Availability	Potential single point of failure	Full proof
Transaction creation	Central authority	Managed by all users

Concept of Blockchain

Block

• Header

- Block Version: The block version number indicates which set of block validation rules to follow. Whenever new rules are introduced, the version number is increased to reference which set of rules were followed.
- Previous Block Hash: It's a 32-bytes field that contains a 256-bits hash (created by SHA-256 cryptographic hashing) of the previous block. This helps to create a linear chain of blocks.
- Merkle Tree Root: A 32-bytes field containing a 256-bit root hash. It's constructed hierarchically combining hashes of the individual transactions in a block.
- Block Type: Public / Private / Consortium
- Timestamp: It's the digitally recorded moment of time when the block has been mined. It is used to validate the transactions.
- Difficulty level: It's a 4-bytes field that tells the complexity to add the block. It's also known as "difficulty bits." According to PoW, the block hash should be less than the difficulty level.
- Nonce: It's a 4-bytes field that contains a 32-bit number. These are the only changeable element in a block of transactions. In PoW, miners alter nonce until they find the right block hash.

Concept of Blockchain

Block

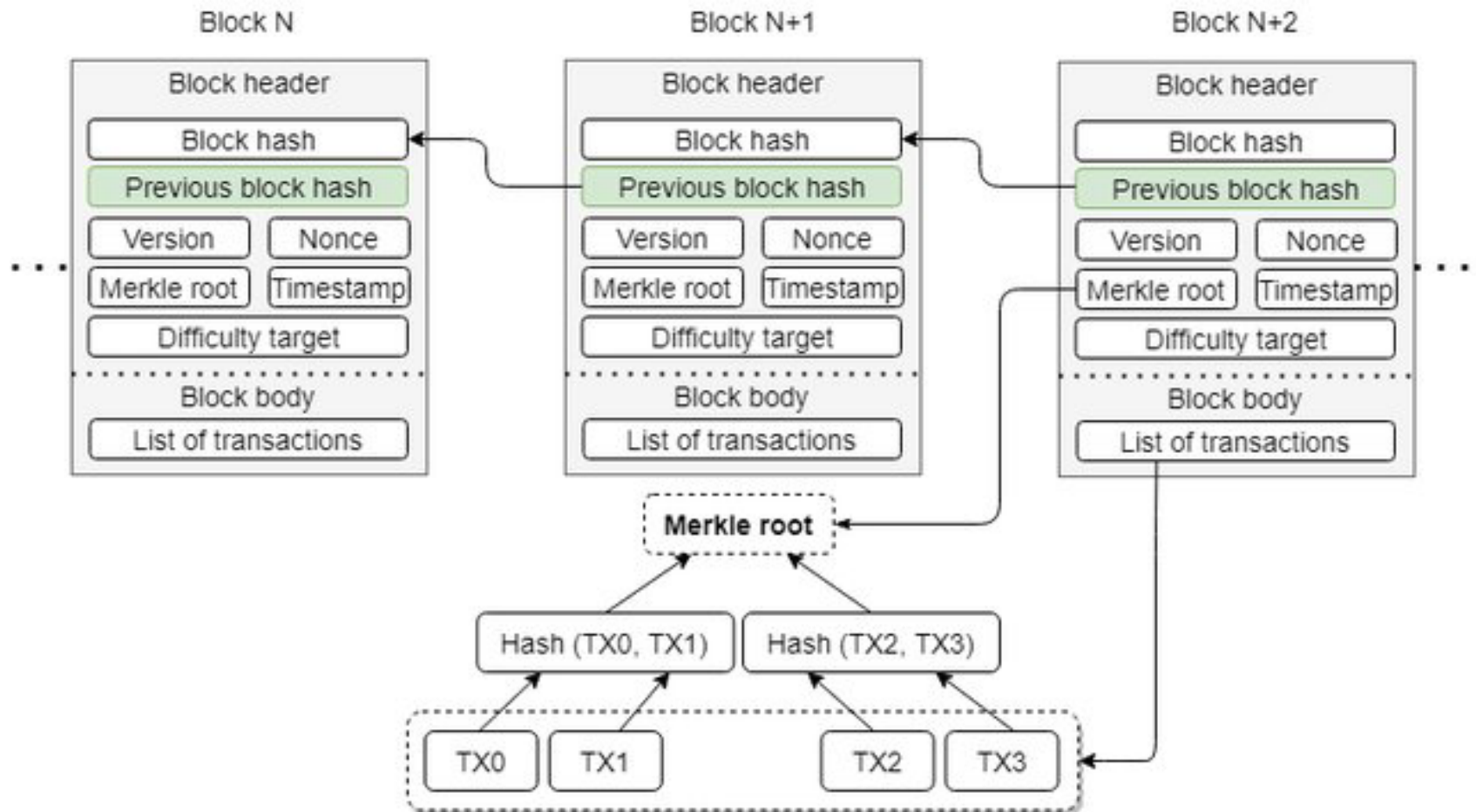
• **Header**

- Block Height: It's the sequence number of the block in the chain of blocks. Block Height: 1 is the genesis block (first block in the network).
- Block Size: It's a 4-bytes or 32-bit field that contains the size of the block. It adds size in Bytes. Ex – Block Size: 216 Bytes.
- Tx Count: The transaction counter shows the number of transactions contained by the block. The field has a maximum size of 9 bytes.

• **Body**

- Transactions: It's a variable-size field that includes the list of all transactions contained in the block.
 - E.g.: Each bitcoin block contains about 2000 transactions.
 - The size of each block is approx 1MB.
 - The size and number of transactions in a block vary in blockchains. It's decided based on network congestion and communication overhead.

Concept of Blockchain



Concept of Blockchain

BLOCK 15152

✓

PREVIOUS BLOCK HASH

00092d17dc

TIMESTAMP

Jul 15, 2019 7:00:29 PM

FROM	TO	AMOUNT
Sylvia	Felicity	\$76.53
Elisabeth	Annabelle	\$24.23
Taylor	Natalie	\$181.90
Ellen	Jakayla	\$302.51
Ali	Salma	\$475.23
Daphne	Lauren	\$127.03
Emilie	Evelin	\$4.05

NONCE

1123

THE BLOCK'S HASH :

000C96E4BC

Concept of Blockchain

```
Block {
  timestamp: '1635473862672',
  data: [],
  prevHash: '',
  hash: 'f8e75f2e328aa77333d14aa3dea8f6c14f750e77e0c1608a6039e41d77ca483b',
  nonce: 0
},
Block {
  timestamp: '1635473862673',
  data: [ [Object] ],
  prevHash: 'f8e75f2e328aa77333d14aa3dea8f6c14f750e77e0c1608a6039e41d77ca483b',
  hash: '0e2be22a826e87a69dcfc9b2ee223c4cc910f7b1b430a63265bb3e3b130761',
  nonce: 0
}
]
```

Concept of Blockchain

Transactions

- A transaction refers to a contract, agreement, transfer, or exchange of assets between two or more parties.
- It is data transmission across the network of computers in a blockchain system.
- The network of computers in a blockchain store the transactional data as replicas with the storage typically referred to as a digital ledger.
- **How are transactions verified ?**
 - Initiate a new transaction
 - Broadcast transaction to a global network of peer-to-peer computers (stored in local MemPool)
 - Nodes confirms the validity of the transaction
 - Validated transactions are clustered into blocks and broadcasted (removed from each nodes MemPool)
 - The blocks are chained together
 - Completion of the transaction

Concept of Blockchain

Transactions

- **Hash encryptions**

- Blockchain employs hashing and encryption technology, mainly the SHA256 algorithm, to ensure data security. Therefore, the SHA256 algorithm transmits the transaction details as encrypted information (hash encryption), which gets added to the blockchain post verification. Thanks to the SHA256 algorithm, hash encryption becomes practically impossible to hack.

- **Consensus**

- In blockchain technology, mining means adding transactions to the distributed digital public ledger of existing transactions (or the blockchain). Although primarily associated with bitcoin, mining also applies to other blockchain usage scenarios. The mining process involves generating a hash of a block of transactions. Since the hash is unforgeable, it protects the integrity of the entire blockchain without requiring a central system such as a bank or regulatory body.

Concept of Blockchain

Hashes

- A hash is generated via a cryptographic function that compares an input block of data with a previously generated hash value.
- If the generated hash value matches the given data, it is determined that the input information hasn't been altered.
- The information contained in the block is hashed, and then the hash value is further added to the end of the chain.
- Any time the data contained in the block is altered or changed in any way, the hash value will also change.
- Properties of a hash function:
 - Deterministic (fixed Input /Output)
 - Quick Computation
 - Collision Resistance

Concept of Blockchain

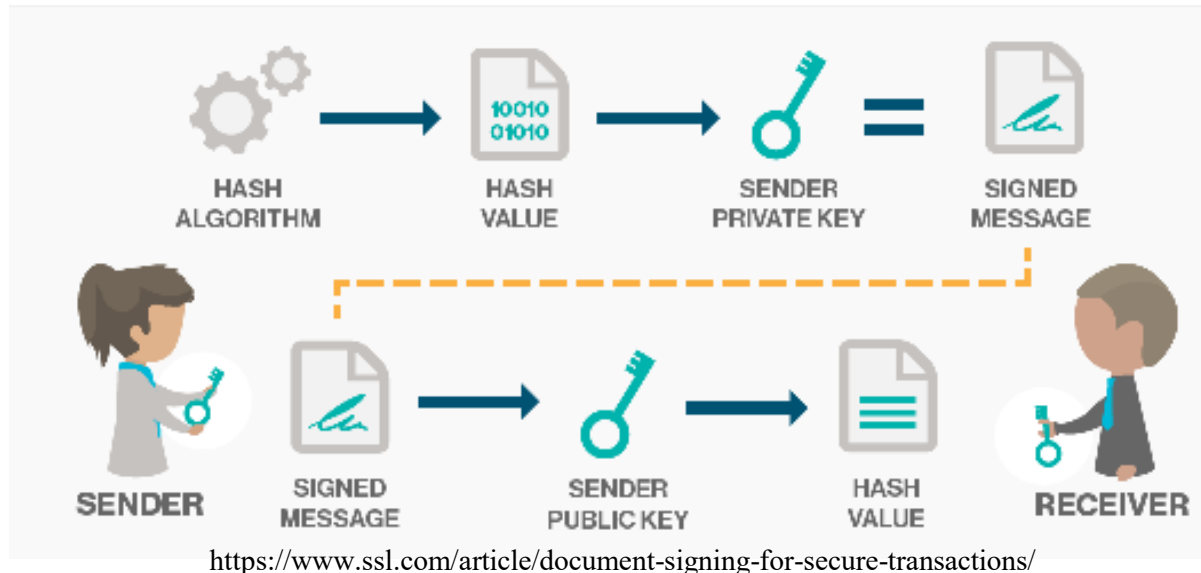
Hashes

- Use of hash in blockchain
 - Generating addresses in blockchain: Public / Private key generation, wallet
 - Determining cryptographic signatures that validate transactions: Public / Private key pair
 - In mining: Nonce, Difficulty level and hashing
 - Storing hash of data makes a block less bulky
- Hashing vs. Encryption
 - Reversibility
 - Integrity vs. Security
 - Length of Output
- There are many different hashing algorithms used to create a hash value.
 - SHA-256 (Secure Hashing Algorithm)
 - MD5 – 128 (Message Digest)
 - RIPEMD-160 (RACE Integrity Primitives Evaluation Message Digest)
 - CRC32 (Cyclic Redundancy Check)
 - Tiger Algorithm – 192
 - Whirlpool - 512

Concept of Blockchain

Transactions

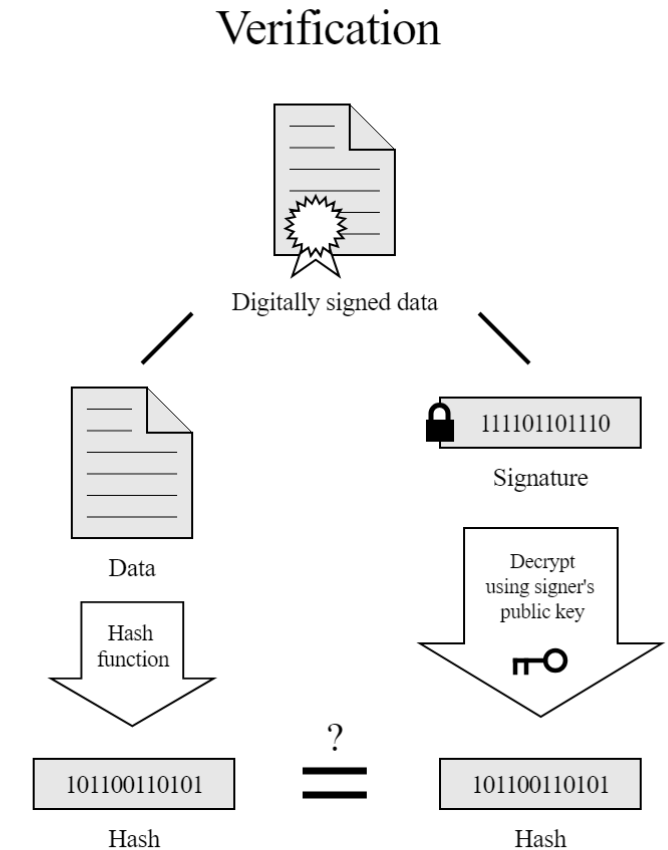
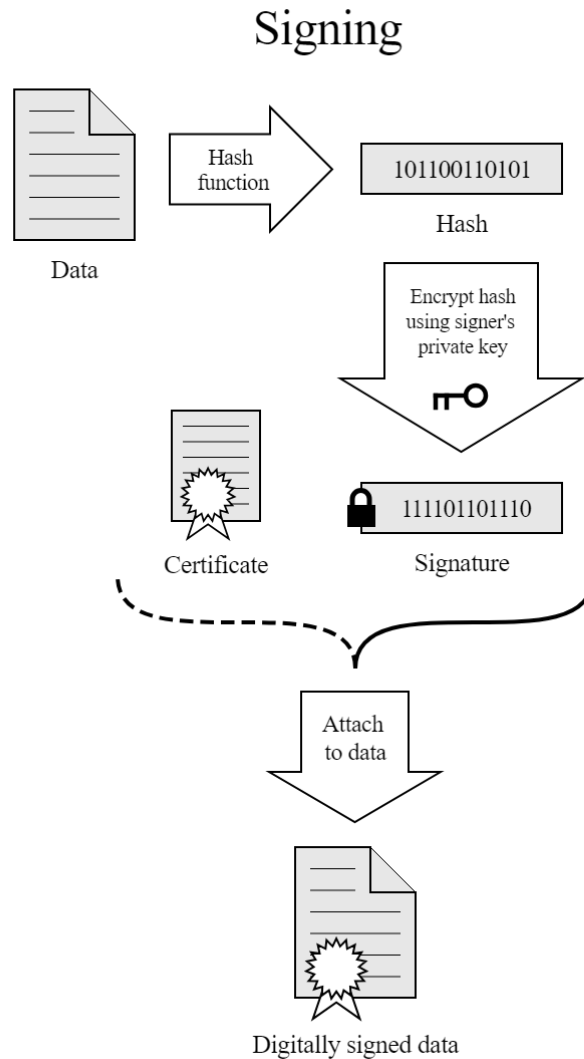
- Transaction is a piece of information that needs to be shared.



- User identity and transaction verification: Anonymity vs. Identification
 - Domain Name System (DNS)
 - Seed Servers
 - Network Message Protocol
 - Peer Discovery

Concept of Blockchain

Transactions

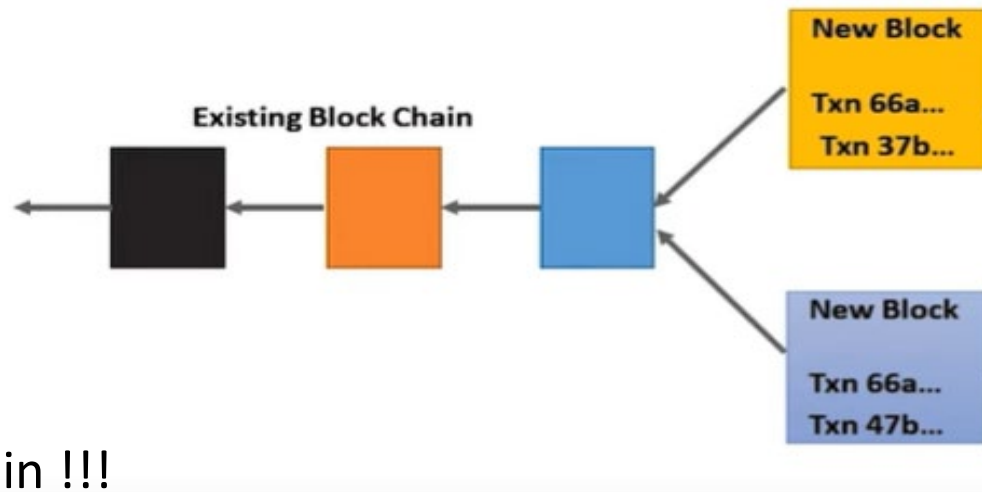


If the hashes are equal, the signature is valid.

Concept of Blockchain

Hashes

- Miners are special nodes which hold the copy of the ledger and verify transactions.
- Miners must be able to
 - Validate the nodes initiating the transaction by means of the public key
 - Originator owns the asset to be transferred.
- Many miners can generate the new block. Which one is added to the blockchain ?

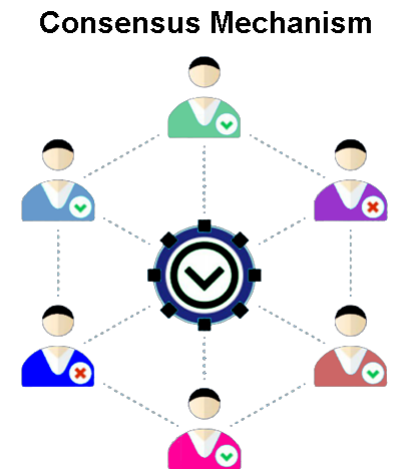


- Consensus comes in !!!

Concept of Blockchain

Consensus Building

- Miners are special nodes which hold the copy of the ledger and verify transactions.
- A consensus mechanism is a system that validates a transaction and marks it as authentic.
- This mechanism ensures that a network of random people agree on the same set of transactions on the network.
- It secures the blockchain's transactions by preventing **double spending problem**.
 - Spending digital asset more than once.
 - Timestamp



Concept of Blockchain

Types of Consensus Mechanism

Proof of Work (PoW): A PoW is data that is difficult to produce but easy for others to verify and also satisfies certain requirements.

Proof of Stake (PoS): It states that a person can mine or validate the block transactions according to how many coins (assets) it holds.

Proof of Authority (PoA): It is used when participants are known to each other and there is a level of trust between them.

Proof of Burn (PoB): It is an alternative consensus algorithm that tries to address the high energy consumption issue of a PoW system. PoB is often called a PoW system without energy waste.

Byzantine Fault Tolerance (BFT): Ability of a network to reach a consensus despite malicious nodes.

Concept of Blockchain

Proof of Work (PoW)

- It is a type of consensus mechanism where the miners use their computational devices to solve complex mathematical problems in order to verify and add blocks.
- It needs heavy computational power to function properly.
- The complexity of these puzzles depends on the number of miners participating in the network and the total computing power they collectively possess, known as the network's **hash rate**.
- The mining **Difficulty** is a measure of how hard it is for miners to solve mathematical puzzles under the PoW mechanism.
- This difficulty level ensures the network's security and stability by regulating the rate at which new blocks are added to the blockchain.
- The more miners there are in the network with high hash rates, the higher the difficulty level will be.

Concept of Blockchain

Proof of Work (PoW)

- To maintain network stability, the mining difficulty is periodically adjusted based on the total network hash rate. This adjustment ensures that the average time between blocks remains relatively constant, regardless of changes in the network's hash rate.
- When more miners join the network, the hash rate increases and the difficulty level rises to keep the block generation time constant.
- If many miners leave the network, lowering the hash rate, the difficulty decreases.
- A difficulty of 1,00,000 means it takes 1,00,000 times more tries (on average) to find a valid hash than with the easiest difficulty of 1. With a constant hash rate in the network, this would lead to a 1,00,000 times longer block finding time.
- **Block time** defines the time it takes to mine a block.
- In Bitcoin, the expected block time is 10 minutes, while in Ethereum it is between 10 to 19 seconds.

Concept of Blockchain

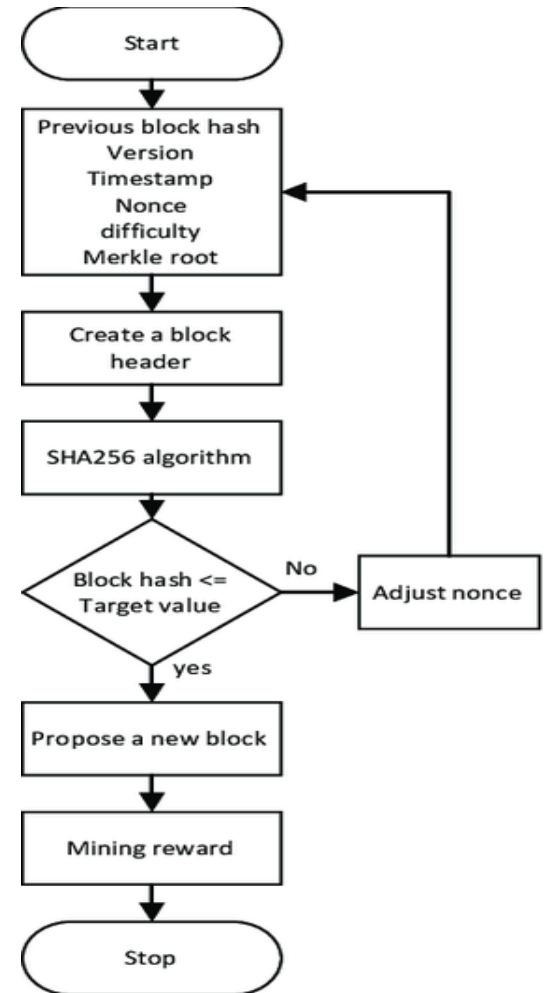
Proof of Work (PoW)

- The average block time of the network is evaluated after n number of blocks;
 - ✓ If it is greater than the expected block time, then the difficulty level of the proof of work algorithm will be reduced.
 - ✓ If it is less than the expected block time then the difficulty level will be increased.
- Example:
 - In Bitcoins, the expectation is to mine a block in 10 minutes.
 - If the average speed of mining the last n (2016) blocks is 8 minutes, then the current difficulty level will be increased.
 - If the average speed of mining is above 10 minutes, then the difficulty level will be decreased for the next 2016 blocks.
 - The difficulty level is reevaluated after every 2016 blocks, that's roughly after every 2 weeks.

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Proof of Work (PoW)

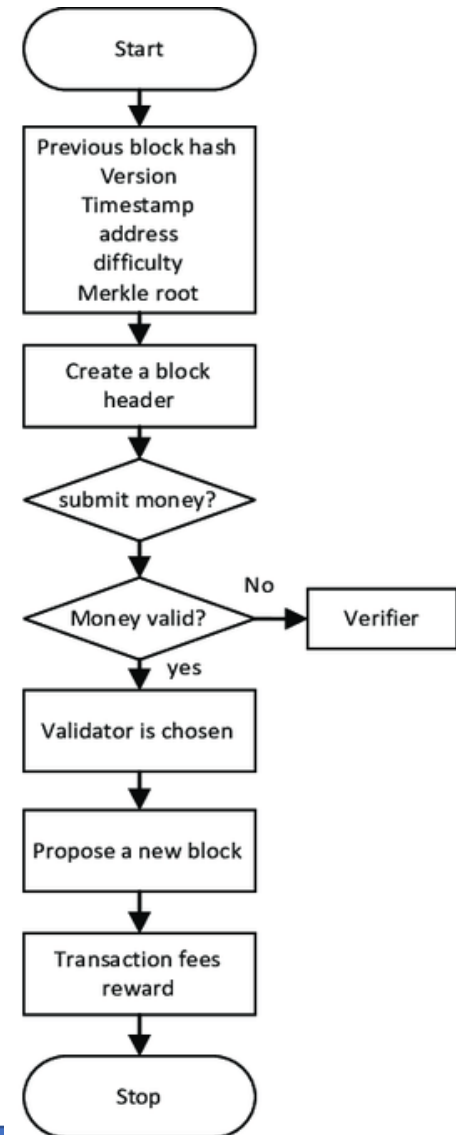
- Advantages:
 - PoW offers a high level of security as compared to other consensus mechanism.
 - Offers transparent transaction verifying process.
 - Also offers rewards for users performing mining.
- Disadvantages:
 - Computationally expensive.
 - Vulnerable to 51% attack.
 - A 51% attack is an attack on a blockchain network where a single entity gains control of more than half (51%) of its staking or computational power.
 - This disproportionate control allows them to implement substantial changes, contravening the decentralization principle fundamental to the blockchain.
 - Large blockchain networks have a low likelihood of a 51% attack.



Concept of Blockchain

Proof of Stake (PoS)

- It is a type of consensus mechanism that depend on a validator's economic stake in a network.
- While PoW mechanisms require miners to solve cryptographic puzzles, PoS mechanisms require validators to hold and stake tokens for the privilege of earning transaction fees.
- For instance, Ethereum requires 32 ETH to be staked before a user can operate a node.
- Validators win transaction fees.
- Slashing
- Validators are chosen based on:
 - The number of staked coins they have
 - How long they have been staking
 - Random



Concept of Blockchain

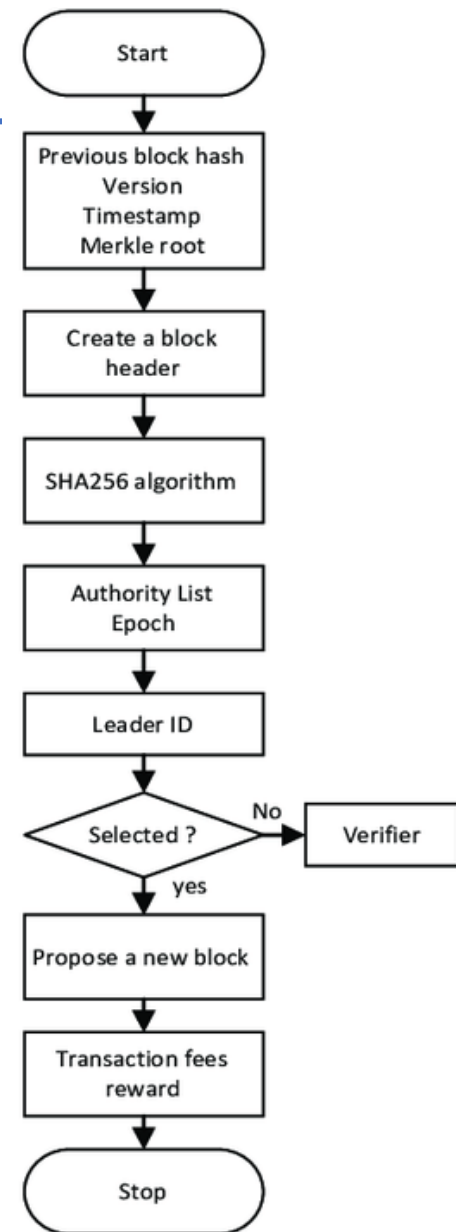
Proof of Stake (PoS)

- Advantages:
 - Energy Efficient
 - Less computational power required
- Disadvantages:
 - Validators who hold large amounts of a blockchain's token or cryptocurrency may have excessive influence on a PoS system.

Concept of Blockchain

Proof of Authority (PoA)

- The Proof-Of-Authority (PoA) is a consensus method that gives a small and designated number of blockchain actors the power to validate transactions or interactions with the network and to update its more or less distributed registry.
- Instead of coins, user's stake their reputation.
- The identity of the validators are known (hence contrasting to the some public blockchain's anonymity criteria)
- More suited for Consortium / Private blockchains.
- Advantages:
 - Energy efficient
 - Speed
- Disadvantages:
 - Not completely anonymous



Concept of Blockchain

Proof of Burn (PoB)

- Proof of Burn (PoB) is a consensus mechanism that uses less energy consumption than Proof of Work (PoW).
- In order for miners to win the right to add a new block to the network, they must 'burn' a portion of their tokens by sending them to an unspendable address, known as an eater address.
- When the coin is burnt its availability decreases leading to a potential increase in the value of the coin.
- It is a deflationary idea in which the quantity of currencies reduces over time, increasing deficiency and, as a result, the currency holders' value. Coins that grow their quantity over time, on the other hand, tend to lose value.
- Advantages:
 - Energy efficient
 - Long term commitment
- Disadvantages:
 - Wasted resources

Concept of Blockchain

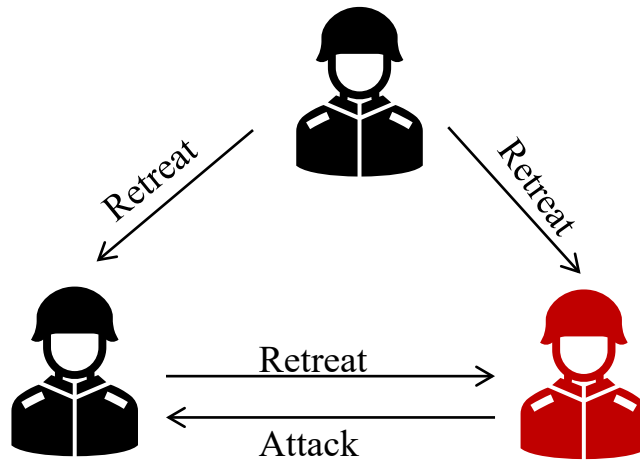
Byzantine Generals Problem

- It is a common problem in distributed computing.
- “Several divisions of the Byzantine army are camped outside an enemy city, each division commanded by its own general. The generals can communicate with each other only by a messenger. After observing the enemy, they must decide upon a common plan of action”
- It is basically a game theory problem that provides a description of the extent to which decentralized parties experience difficulties in reaching consensus without any trusted central parties.
- The Byzantine army is divided into many battalions in this classic problem called the Byzantine General’s problem, with each division led by a general.
- The generals connect via messenger in order to agree to a joint plan of action in which all battalions coordinate and attack from all sides in order to achieve success.
- Generals should reach a common consensus as to Attack or Retreat !

Concept of Blockchain

Byzantine Generals Problem

- It is probable that traitors will try to sabotage their plan by intercepting or changing the messages.
- As a result, the purpose of this challenge is for all of the faithful commanders to reach an agreement without the imposters tampering with their plans.
- 3 Byzantine Problem



Concept of Blockchain

Byzantine Fault Tolerant (BFT)

- Byzantine Fault Tolerance is a computer system's ability to continue operating even if some of its nodes fail or act maliciously.
- More than $1/3^{\text{rd}}$ of the node cannot be faulty (33%)
- BFT means that in spite of Byzantine (traitor/faulty) nodes, the network should be able to function properly.
- Primary / Secondary nodes
- The primary (leader) node is responsible for proposing new transactions or blocks to the network.
- The secondary (replica) nodes are responsible for validating the proposal by exchanging messages with each other.
- If two-thirds of the secondary nodes agree on the proposal, it is considered to be validated, and the leader node adds it to the blockchain.

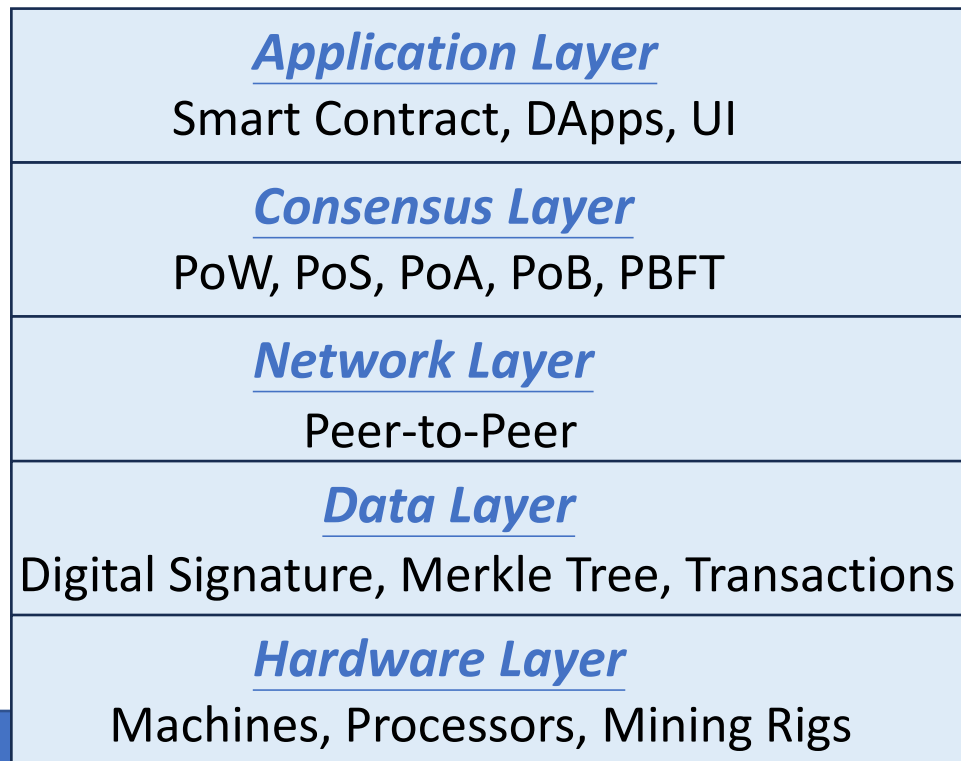
Concept of Blockchain

Byzantine Fault Tolerant (BFT)

- It doesn't require significant computing power or energy usage because there aren't any miners solving complex equations for every block of transactions. This makes it much more environmentally friendly than proof of work.
- Since all nodes can get in on the action, they can all share in the rewards. There isn't such variance between which nodes earn rewards like there can be in proof of work and proof of stake.
- It requires a high level of network connectivity between nodes. If nodes are not able to communicate with each other quickly and reliably, consensus may not be reached, and the network may become fragmented.

Blockchain Architecture

- Blockchain architecture refers to the components and subcomponents or layers that make up a complete blockchain system.
- Each layer has a unique role, from storing data to maintaining network connectivity and ensuring consensus across the system.
- Depending on a blockchain type, there are different ways to implement its architecture, and there is no standard for it. A common architecture includes:



Blockchain Architecture

Application Layer

- Provides blockchain-based applications for the end user.
- The application layer starts with a smart contract which is a programmable code.
- Usually, users don't interact with the smart contract directly. Instead, they rely on a front end of a Web3 application or an API.
- Metamask

Consensus Layer

- Consensus ensures agreement among nodes for block mining and processing and outlines validator requirements, varying across proof-of-work, proof-of-stake, and other consensus mechanisms.

Blockchain Architecture

Network Layer

- The network layer enables effective discovery and interaction among nodes.
- Typically, a node locates a primary / bootnode, which scans for available peers and initiates bonding.

Data Layer

- The data layer of blockchain technology is primarily concerned with data storage and structure.
- Depending on the specific blockchain, the data structure can range from a simple transaction list to tries.
- Maintains data integrity and privacy.
- Digital signature algorithms, cryptographic libraries, public-private key pairs

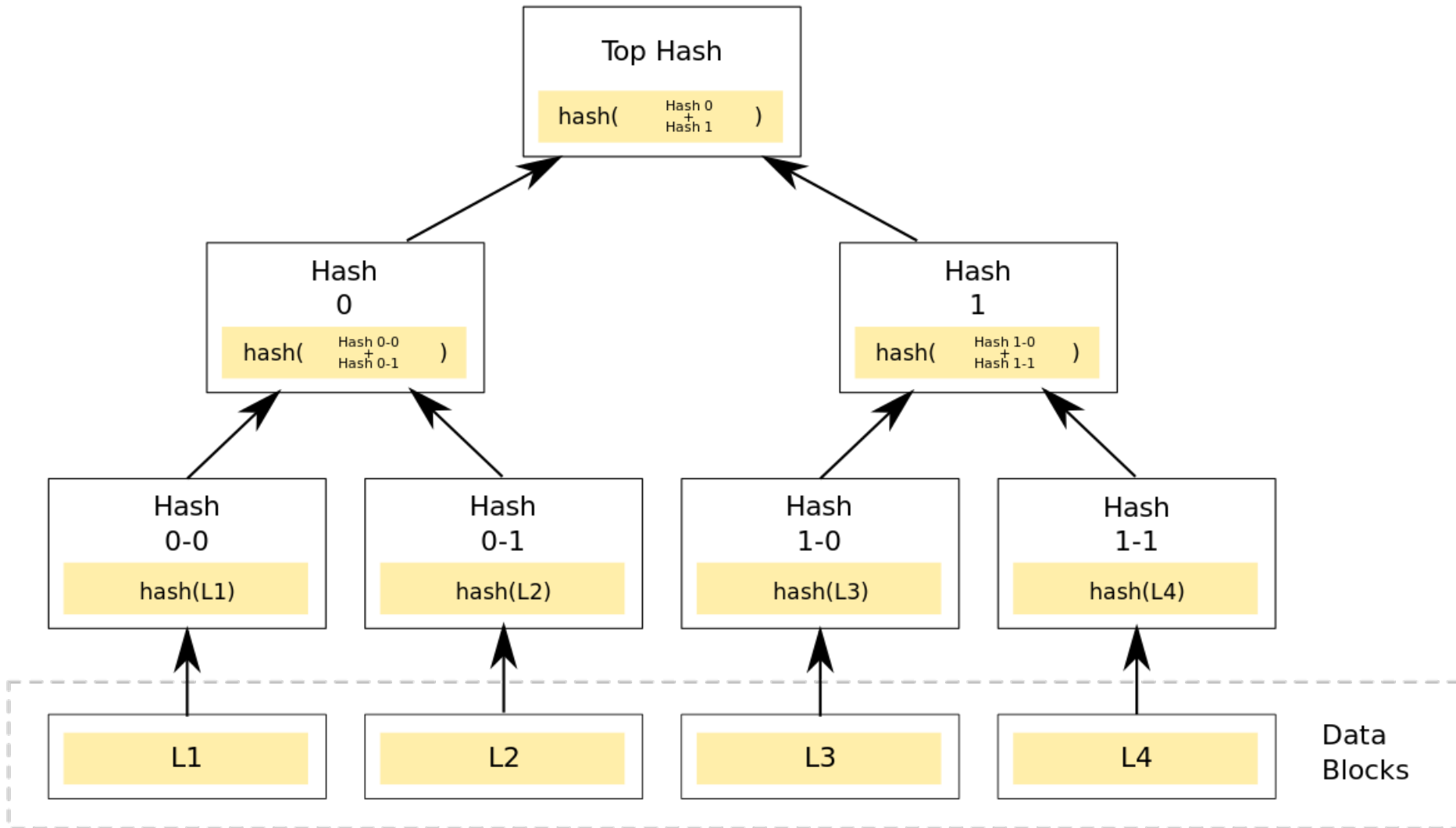
Hardware/Infrastructure Layer

- Full nodes / Partial
- Machines / Processors / Mining Rigs

Merkle Tree

- A hash tree or Merkle tree is a tree in which every "leaf" (node) is labelled with the cryptographic hash of a data block, and every node that is not a leaf (called a branch, inner node) is labelled with the cryptographic hash of the labels of its child nodes.
- This structure of the tree allows efficient mapping of huge data and small changes made to the data can be easily identified.
- If we want to know where data change has occurred then we can check if data is consistent with root hash and we will not have to traverse the whole structure but only a small part of the structure.
- The root hash is used as the fingerprint for the entire data.
- If there is an odd number of transactions, one transaction is doubled and its hash is concatenated with itself.

Merkle Tree



https://en.wikipedia.org/wiki/Merkle_tree